

Karan Singh Sodhi – Software Engineer

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Summary

Entry-level Software Engineer with hands-on experience building **full-stack and real-time web applications** using MERN and Next.js. Strong foundation in **data structures, RESTful API design, object-oriented programming, and backend performance optimization**. Experienced with **Docker**, and real-time systems using Socket.IO through internships and production-grade projects. Comfortable working in agile environments, debugging complex issues, and delivering reliable software.

Experience

Web Developer Intern, V'NSPIRE – Agra, India

- Shipped a **production-ready MERN application** with authentication, posting, and feed workflows, handling **50+ user-generated posts**.
- Delivered **real-time messaging** using Socket.IO, supporting **10+ concurrent users** with low-latency updates.
- Modeled and optimized **MongoDB collections** for users, posts, and messages, scaling to **1,000+ records** without performance degradation.
- Designed and exposed **10+ REST APIs** in Node.js and Express with clear contracts, validation, and error handling.
- Secured backend services using **JWT-based auth** and role checks, preventing unauthorized access to protected endpoints.
- Improved UI responsiveness and data flow through efficient client-side state management and render optimization.

Projects

Real-Time Collaboration Platform (Discord Clone) *Next.js 15, Socket.IO, LiveKit, PostgreSQL* [Live](#) | [Code](#)

- Engineered a scalable real-time communication system supporting low-latency messaging and voice/video streaming using WebSockets and SFU-based architecture.
- Designed normalized relational schemas with indexing, cascading constraints, and soft-deletion strategies to ensure data integrity and efficient querying.
- Implemented role-based access control, secure authentication flows, and server-side validation for isolated multi-server collaboration.
- Optimized event handling and state synchronization to minimize re-renders and reduce client-side performance overhead.

Scalable E-Commerce Platform (E-Kart) *MongoDB, Express.js, React.js, Node.js, Stripe, Cloudinary* [Live](#) | [Code](#)

- Architected and deployed a production-ready full-stack eCommerce platform with modular REST APIs and scalable MERN architecture.
- Designed secure JWT-based authentication and role-based authorization (Admin/User), enforcing protected routes and access isolation.
- Integrated Stripe payment workflows with end-to-end order validation and Cloudinary-backed media optimization for efficient asset delivery.
- Optimized database indexing and query patterns in MongoDB, reducing response latency and improving overall application performance.

Education

Dr. A.P.J. Abdul Kalam Technical University

08/2023 – 08/2027

Bachelor of Technology in Computer Science and Engineering

Skills

Languages: C, C++, Java, Python, JavaScript

Frontend: React.js, Next.js, HTML, CSS

Backend: Node.js, Express.js, REST APIs, Asynchronous Processing, Authentication

Databases: PostgreSQL, MongoDB, Indexing, Query Optimization

Tools: Git, GitHub, Debugging, Problem Solving

CS Fundamentals: OOP, Git, GitHub, Data Structures, Algorithms, Operating Systems, DevOps

Certifications

- Full Stack Web Development (MERN)** — Sheriyans Coding School
- JavaScript (ES6+) & Data Structures** — Coursera
- AWS Cloud Practitioner & Fundamentals** — freeCodeCamp